

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)
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≤ Academic Performance Prediction Using Machine Learning ≥

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BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by:

Chirag Dahiya	2210990239
Anmol Kumar	2210990122
Augustya	2210990196
Aryan Sonkar	2210990182

Supervised By:

Dr.Gifty Gupta
Assistant Professor
Department of Computer science
Chitkara University



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY
CHITKARA UNIVERSITY, PUNJAB, INDIA

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Academic Performance Prediction Using Machine Learning**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr.Gifty Gupta** The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Date: 14 may,2026

Chirag Dahiya

(University Roll No 2210990239)

Anmo.Kumar . . .

(University Roll No 2210990122)

Augustya . . .

(University Roll No 2210990196)

Aryan..Sonkar .

University Roll No.2210990182

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

(Dr.Gifty Gupta)

Assistant Professor

Department of Computer Science and Engineering

Chitkara University Institute of Engineering and Technology,

Chitkara University, Punjab, India

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ABSTRACT

In the modern education system, identifying students who are at risk of poor academic performance is a critical challenge. Traditional evaluation methods rely heavily on final examinations, which often fail to provide timely insights into a student's learning progress. This delay makes it difficult for educators to intervene effectively.

This project presents a machine learning-based approach to predict student academic performance using multiple influencing factors such as attendance, internal assessment scores, study hours, and past academic records. The system leverages supervised learning algorithms including Logistic Regression, Decision Tree, and Random Forest to analyze patterns in student data and generate predictive outcomes.

Data preprocessing techniques such as normalization, handling missing values, and feature selection are applied to improve model efficiency. The dataset is divided into training and testing subsets to evaluate model performance using metrics like accuracy, precision, recall, and confusion matrix.

The results demonstrate that the Random Forest algorithm outperforms other models with the highest accuracy, making it the most suitable approach for this prediction system. The proposed system enables early identification of weak students and supports data-driven decision-making in educational institutions.

Keywords: Machine Learning, Student Performance Prediction, Random Forest, Educational Data Mining, Classification

1.INTRODUCTION

Education is one of the most important pillars of societal development. The performance of students directly reflects the effectiveness of teaching methodologies and learning environments. However, traditional systems of evaluating student performance rely primarily on final examinations, which only provide a retrospective view of student success or failure.

Such systems lack the ability to monitor continuous progress and fail to identify students who may be struggling during the learning process. As a result, educators often miss opportunities to provide timely support and intervention.

With the advancement of technology, particularly in the field of machine learning, it is now possible to analyze large datasets and identify hidden patterns that can help predict future outcomes. Machine learning algorithms can process historical student data and generate predictive models that estimate academic performance before final exams.

This project aims to develop a predictive system using machine learning techniques to analyze student data and forecast their academic performance. The system not only improves prediction accuracy but also helps educators take proactive steps to improve student outcomes.

1.1 Background

The rapid growth of digital education, especially after the COVID-19 pandemic, has significantly increased the availability of student data. Educational institutions now have access to detailed records such as attendance, assignment scores, participation levels, and exam results. many institutions still rely on traditional evaluation techniques that do not fully utilize this information. Existing e-learning systems often lack predictive capabilities and fail to provide actionable insights.

Common issues in current systems include:

- Lack of real-time monitoring

- Fragmented data across multiple platforms
- Absence of intelligent analytics
- Limited support for personalized learning

Educational Data Mining (EDM) and Machine Learning (ML) have emerged as powerful tools to address these challenges. These technologies enable the extraction of meaningful patterns from large datasets, helping institutions make informed decisions.

This project builds upon these advancements to create a predictive model that enhances student performance analysis.

1.2. Problem Statement

Despite technological advancements, current academic systems face several limitations:

- Delayed Performance Evaluation : Students are evaluated only after final exams, leaving no scope for early intervention.
- Lack of Predictive Systems: Most institutions do not use data-driven approaches to predict student outcomes.
- Inefficient Resource Utilization: Teachers cannot focus on students who need the most attention due to lack of insights.
- Limited Consideration of Factors : Academic performance depends on multiple variables, but traditional systems consider only exam results.
- No Personalization : Learning strategies are not adapted based on individual student needs.

This project addresses these issues by implementing a machine learning-based prediction system.

1.3. SOFTWARE AND HARDWARE REQUIREMENTS

This section outlines the software tools and hardware configuration required for the development and execution of the proposed student performance prediction system.

1.3.1. Software Requirements

The implementation of the system is carried out using the Python programming language due to its simplicity, flexibility, and strong support for data analysis and machine learning.

The following libraries are utilized:

- Pandas: Used for data manipulation and preprocessing, including handling datasets, cleaning data, and performing exploratory analysis.
- NumPy: Provides support for numerical computations and efficient handling of arrays and mathematical operations.
- Scikit-learn: A widely used machine learning library that offers various classification algorithms such as Logistic Regression, Decision Tree, and Random Forest, along with tools for model training and evaluation.

Development Tools:

- Jupyter Notebook: Provides an interactive environment for writing and executing code, visualizing outputs, and documenting the workflow.
- Visual Studio Code (VS Code): Used as an alternative integrated development environment (IDE) for coding, debugging, and project management.

1.3.2. Hardware Requirements

The system does not require high-end hardware and can be implemented on standard computing devices. Minimum requirements are ram 4gb, processor intel i3 , storage 256 gb.

2. METHODOLOGY

The methodology adopted in this study focuses on developing and evaluating a classification model using structured data. It consists of four main stages: data preparation, model selection, training, and testing, ensuring accuracy and reliability of results.

2.1. Data Preparation

Data preparation involves converting raw data into a clean and usable format. The dataset is first examined for missing values, inconsistencies, and noise. Missing data is handled using imputation techniques or by removing incomplete records when necessary, while outliers are identified and treated.

The data is then structured appropriately. Categorical variables are encoded using methods such as label encoding or one-hot encoding, and numerical features are normalized to maintain uniformity. Relevant features are selected to improve model performance and reduce complexity. Finally, the dataset is split into training and testing sets, typically in an 80:20 ratio.

2.2. Model Selection

In this stage, suitable machine learning algorithms are chosen based on the classification problem. Common models considered include Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and K-Nearest Neighbors.

The selection is based on factors such as accuracy, efficiency, and interpretability. Multiple models may be compared, and hyperparameters are tuned using techniques like grid search to identify the best-performing model.

2.3. Training Phase

During training, the selected model is trained using the training dataset. The algorithm learns relationships between input features and target outputs by optimizing a loss function.

Techniques such as cross-validation and regularization are used to prevent overfitting and improve generalization. Performance metrics like accuracy, precision, recall, and F1-score are monitored to assess model learning.

2.4. Testing Phase

The trained model is evaluated using unseen test data to measure its real-world performance. Predictions generated by the model are compared with actual values. Evaluation is carried out using metrics such as accuracy, precision, recall, F1score, and confusion matrix. These results help determine the effectiveness and reliability of the model.

2.5. Experimental Setup

The experimental setup defines the framework used to train and evaluate the machine learning models for predicting student performance.

The dataset used in this study was divided into two subsets: 80% of the data was allocated for training the models, while the remaining 20% was reserved for testing their performance on unseen data. This split ensures that the model is trained on sufficient data while also being evaluated fairly for generalization.

To assess the effectiveness of the models, multiple evaluation metrics were employed. Accuracy was used to measure the overall correctness of predictions. Precision and recall were calculated to evaluate the model's performance in identifying specific classes, especially in cases of class imbalance. The F1-score, which is the harmonic mean of precision and recall, provided a balanced measure of performance. Additionally, a confusion matrix was used to visualize the classification results and identify misclassifications.

2.6. Summary

This methodology provides a systematic approach for building a robust classification model, ensuring efficient data handling, accurate predictions, and reliable evaluation. them, the study aims to achieve high accuracy and robust performance.

3. IMPLEMENTATION

The proposed system follows a structured pipeline to analyze student data and predict academic performance using machine learning techniques. The architecture consists of the following stages:

3.1. Data Collection

Data is collected from student academic records, including key parameters such as attendance, internal marks, study hours, and previous grades. This data forms the input for the system and is obtained from institutional databases or surveys.

3.2. Data Preprocessing

The collected data is cleaned and prepared to ensure accuracy and consistency. Missing values are handled using appropriate techniques such as imputation or removal. Inconsistencies and duplicate records are eliminated. Numerical data is normalized to maintain uniformity and improve model performance.

3.3. Feature Selection

Relevant features are selected to improve the efficiency and accuracy of the model. Unnecessary or redundant attributes are removed, ensuring that only significant factors influencing student performance are retained.

3.4. Model Training

- The system uses three machine learning algorithms for classification:
- Logistic Regression
- Decision Tree
- Random Forest
- These models are trained on the prepared dataset to learn patterns between input features and student performance outcomes.

3.5. Prediction

- After training, the models are used to predict student performance categories such as high, average, or low performer based on input data.

3.6. Evaluation

- The performance of each model is evaluated using metrics such as accuracy, precision, recall, and F1-score. The best-performing model is selected based on these evaluation results.

3.7. Summary

- The system architecture provides a clear and systematic approach for predicting student performance, ensuring reliable and accurate results through effective data processing and model evaluation.

4.RESULTS AND DISCUSSION

The performance of three machine learning algorithms—Logistic Regression, Decision Tree, and Random Forest—was analyzed and compared.

Logistic Regression, being a simple and efficient algorithm, provided faster results but showed relatively lower accuracy compared to other models. The Decision Tree model offered better interpretability, allowing clear understanding of decision paths, and achieved moderate accuracy.

Among all models, the Random Forest algorithm performed the best, achieving an accuracy of approximately 88%. Its ensemble nature, which combines multiple decision trees, helped in improving prediction accuracy, reducing overfitting, and enhancing generalization.

The analysis revealed that attendance and internal marks were the most significant factors influencing student performance. It was also observed that ensemble methods, such as Random Forest, consistently outperformed individual models, making them more suitable for this classification task.

4.1.APPLICATIONS

The proposed system has several important applications in the field of education. It can be used for the early identification of students who are at risk of poor academic performance, allowing educators to take timely corrective measures.

Additionally, the system can support the development of personalized learning strategies tailored to individual student needs. It can also assist institutions in monitoring academic progress, planning interventions, and guiding students in making informed academic and career decisions.

4.2. LIMITATIONS

Despite achieving promising results, the system has certain limitations. The accuracy of predictions is highly dependent on the quality and completeness of the dataset used. Any missing or biased data can affect model performance. Furthermore, the system does not account for emotional, psychological, or social factors that may influence student performance. Another limitation is that complex models like Random Forest, while accurate, may be less interpretable compared to simpler models. Additionally, model training and evaluation require adequate computational resources.

5. FUTURE SCOPE

There is significant scope for improving and extending this system in the future. Advanced techniques such as deep learning can be integrated to further enhance prediction accuracy.

The system can be developed into a real-time prediction platform that continuously analyzes student data. Inclusion of behavioral and psychological parameters could provide a more holistic assessment of student performance. Moreover, the system can be deployed as a web or mobile application for easy accessibility by students and educators. AI-based recommendation systems can also be incorporated to provide personalized suggestions for academic improvement.

6. CONCLUSION

This project successfully demonstrates the application of machine learning techniques in predicting student academic performance. By analyzing key factors such as attendance, internal marks, and study patterns, the system is able to generate accurate and meaningful predictions.

Among the models evaluated, Random Forest proved to be the most effective due to its high accuracy and ability to generalize well. The system promotes a proactive approach to education by enabling early identification of underperforming students and supporting timely interventions.

With further enhancements and real-world integration, this system has the potential to significantly improve educational outcomes and decision-making processes in academic institutions.

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